
EECS 16A Imaging 2

*TA, UCS1, UCS1, UCS1 *

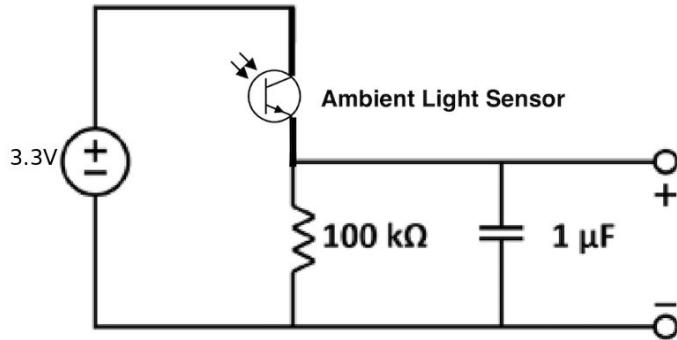
Agenda

- Quick overview + review
- Images as matrices and vectors
- Pixel-by-pixel scanning
- Reconstructing scans as images

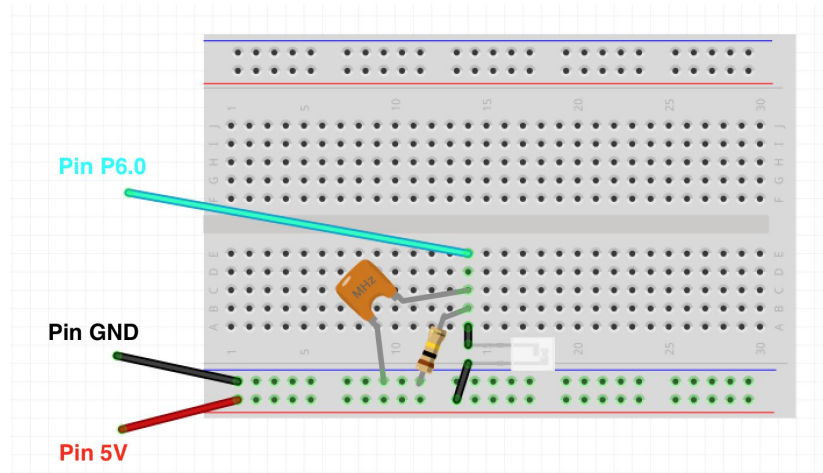
Last Week: Imaging 1

- Built our very first circuit!
 - What did this circuit do?

Circuit Diagram



Breadboard Diagram



Today's Lab: Single Pixel Scanning

- Circuit from last week measures **light** intensity
- Simulated projector illuminates image in a controlled way
- Python programming to reconstruct image

Why?

- Imaging 1:
 - Finding a link between physical quantities and voltage is powerful
 - If you can digitize it, you can do anything (IOT devices, internet, code, processing)
- Imaging 2:
 - How do we get measurements and what makes them good?
 - How do we get to an image?

Illuminating the Big Picture

- Linear dependence
 - When can you recover your image?
 - Does it matter what mask matrix you pick?
 - Does it matter how you cover the pixels?
- Invertibility
 - When can you solve $Ax = b$?
 - How does this relate to our system?
 - How does this affect the way we pick our masking matrix?

Images, Matrices, Vectors



- What are the unknowns in our system?
 - The Image !
- We can do a lot of interesting processing on vectors, but we need to convert the image into one first
 - How can we do this?

Images, Matrices, Vectors



[0]	[1]
[2]	[3]
[4]	[5]

Images, Matrices, Vectors



[0]	[1]
[2]	[3]
[4]	[5]



[0]
[1]

Images, Matrices, Vectors



[0]	[1]
[2]	[3]
[4]	[5]

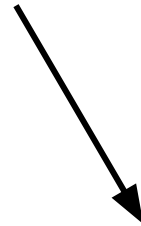


[0]
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Images, Matrices, Vectors



[0]	[1]
[2]	[3]
[4]	[5]



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Images, Matrices, Vectors



[0]	[1]
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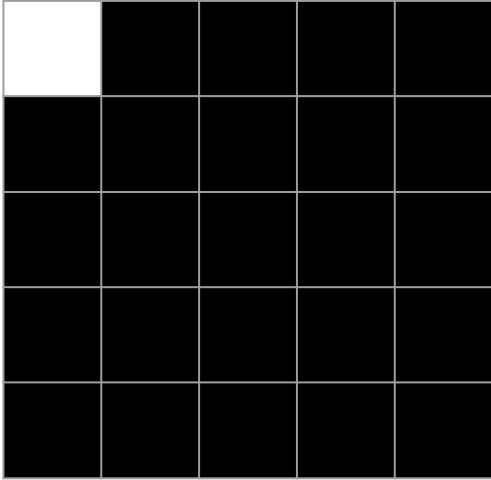


[0]
[1]
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Pixel-by-Pixel Scan of an Image



Pixel-by-Pixel Scan of an Image



Pixel-by-Pixel Scan of an Image

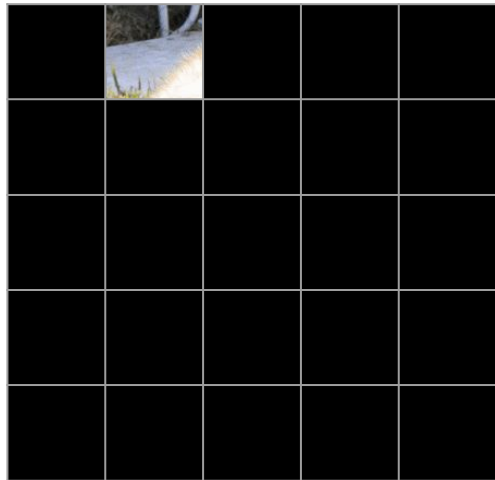


Masked image



Image

Pixel-by-Pixel Scan of an Image

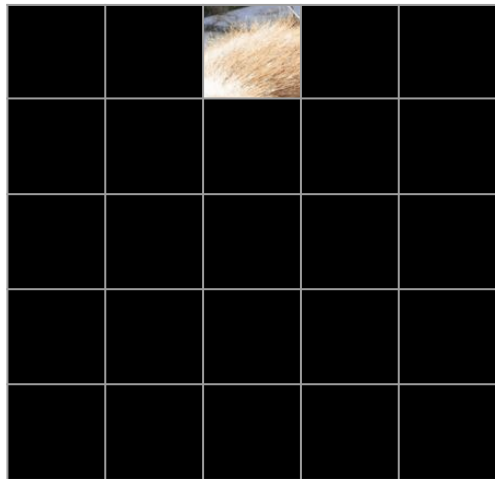


Masked image



Image

Pixel-by-Pixel Scan of an Image



Masked image



Image

Poll Time!

What would you expect the dimensions of a vector representing a 2×3 image to be?

- A. 2×3
- B. 3×2
- C. 6×1
- D. 5×1

To read all the pixels of a 4×4 image, how many pixel-by-pixel scans do we need to do?

- A. 4
- B. 8
- C. 16
- D. 32

Poll Time!

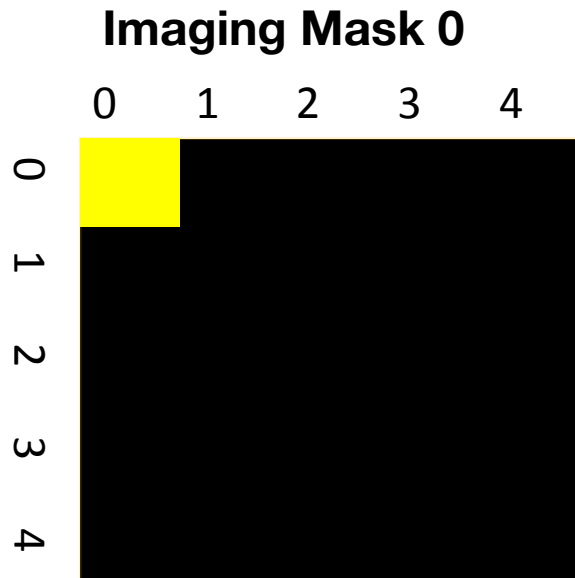
What would you expect the dimensions of a vector representing a 2x3 image to be?

- A. 2x3
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- D. 5x1

To read all the pixels of a 4x4 image, how many pixel-by-pixel scans do we need to do?

- A. 4
- B. 8
- C. 16
- D. 32

Representing our Masks in Python



mask0 =

np.array([[1, 0, 0, 0, 0]
[0, 0, 0, 0, 0]
[0, 0, 0, 0, 0]
[0, 0, 0, 0, 0]
[0, 0, 0, 0, 0]])

Representing our Masks in Python



mask1 =

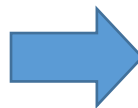
np.array([[0, 1, 0, 0, 0]
[0, 0, 0, 0, 0]
[0, 0, 0, 0, 0]
[0, 0, 0, 0, 0]
[0, 0, 0, 0, 0]])

Turning the Masks Into Vectors

5x5 mask to 25x1 vector

mask0 =

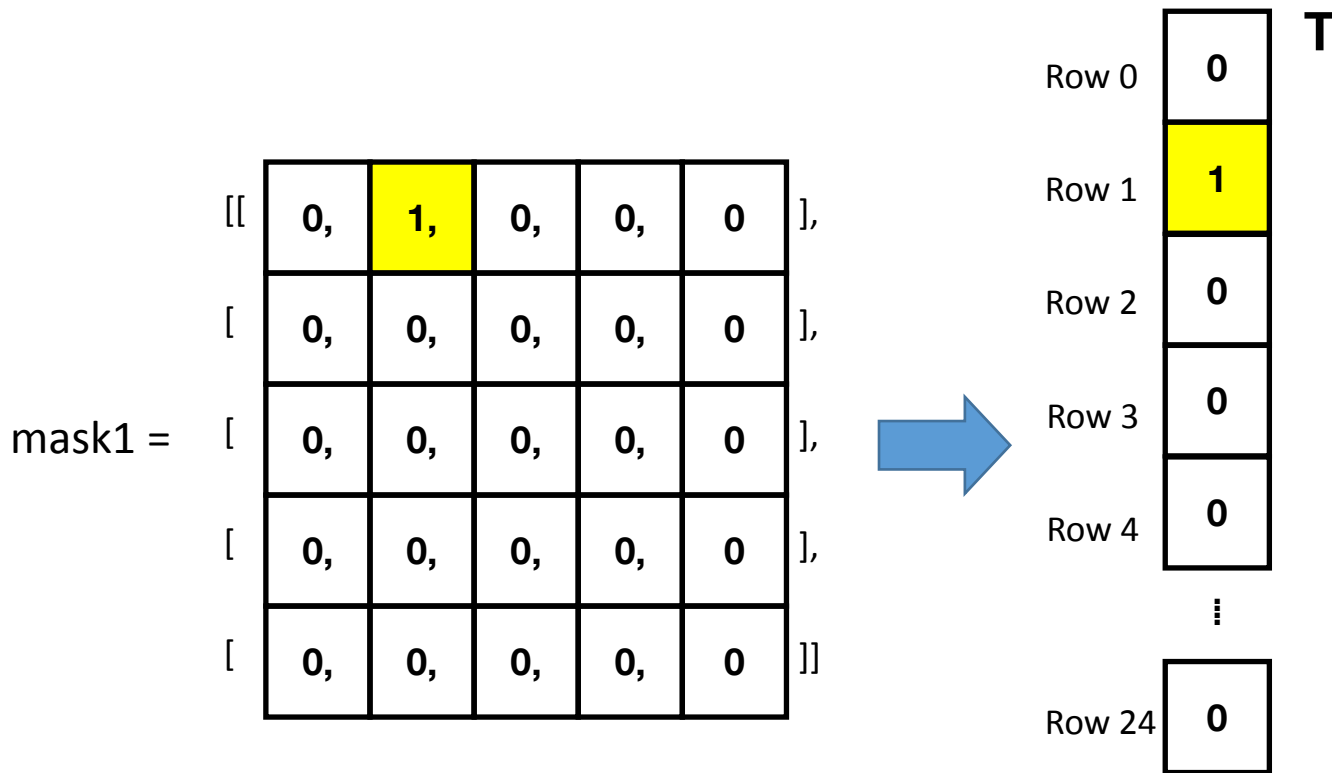
1	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0



Row 0	1
Row 1	0
Row 2	0
Row 3	0
Row 4	0
	⋮
Row 24	0

T

Turning the Masks Into Vectors

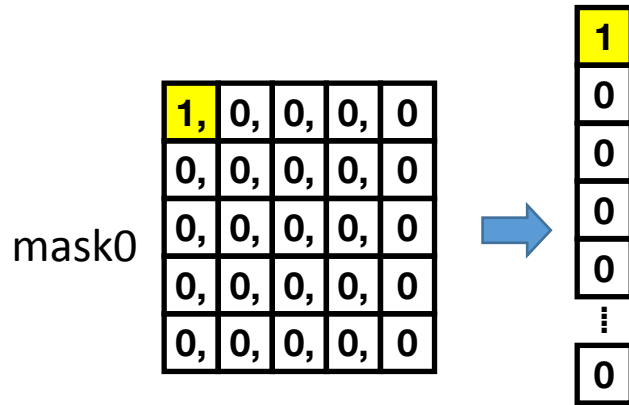


Generating the Masking Matrix from the Masks

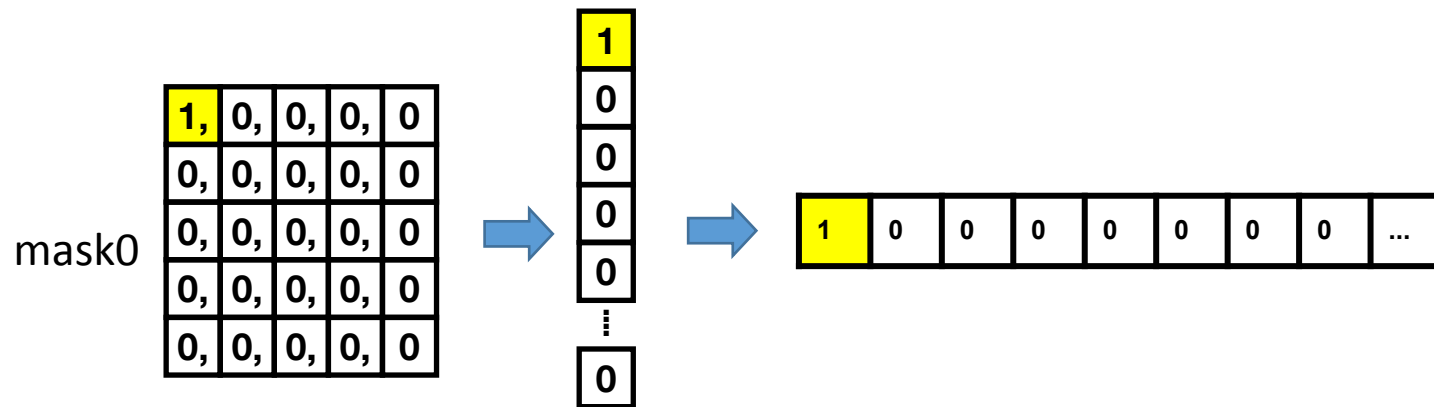
mask0

1,	0,	0,	0,	0
0,	0,	0,	0,	0
0,	0,	0,	0,	0
0,	0,	0,	0,	0
0,	0,	0,	0,	0

Generating the Masking Matrix from the Masks



Generating the Masking Matrix from the Masks



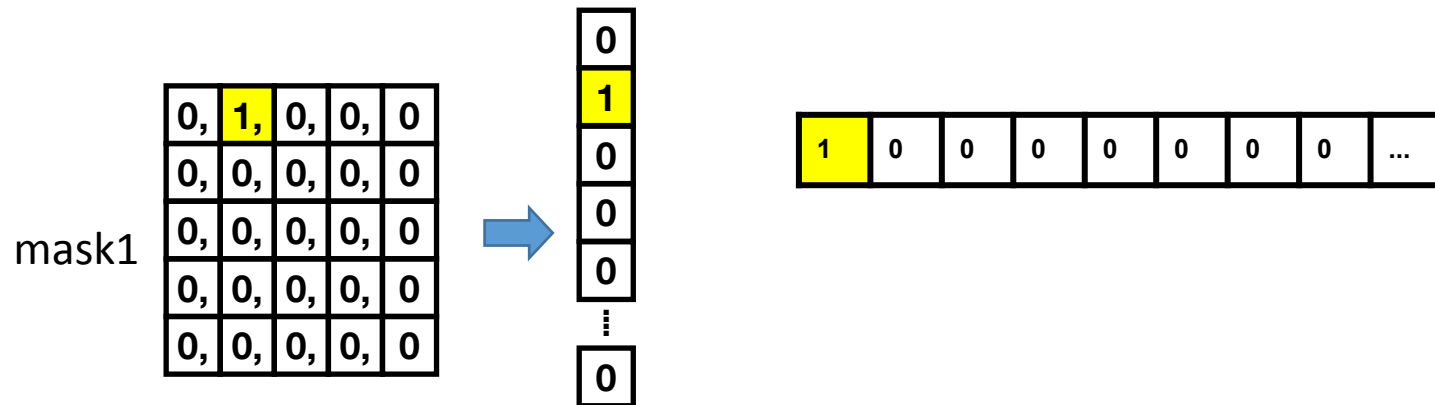
Generating the Masking Matrix from the Masks

mask1

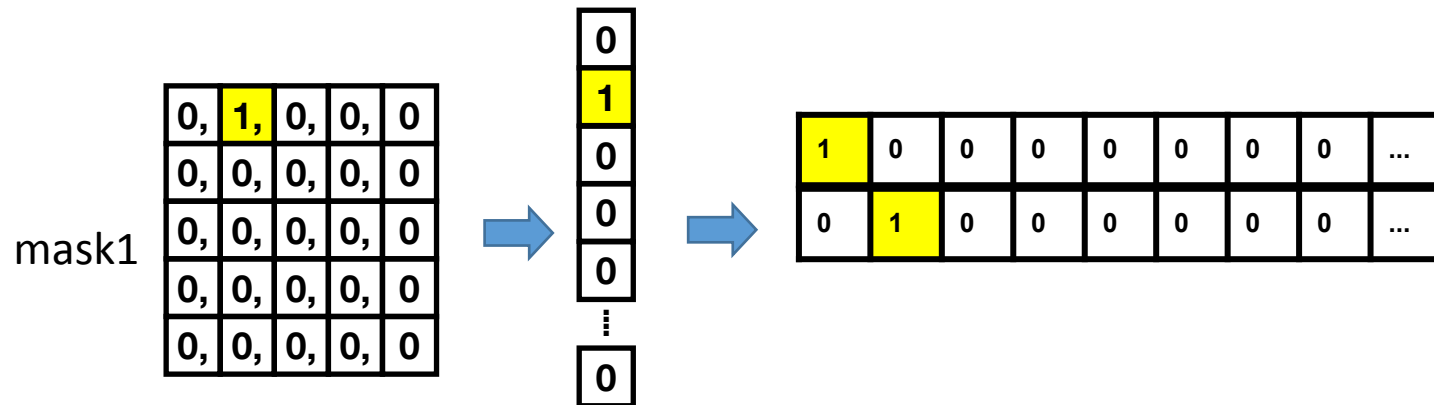
0,	1,	0,	0,	0
0,	0,	0,	0,	0
0,	0,	0,	0,	0
0,	0,	0,	0,	0
0,	0,	0,	0,	0

1	0	0	0	0	0	0	0	...
---	---	---	---	---	---	---	---	-----

Generating the Masking Matrix from the Masks



Generating the Masking Matrix from the Masks



Generating the Masking Matrix from the Masks

1	0	0	0	0	0	0	0	...
0	1	0	0	0	0	0	0	...
0	0	1	0	0	0	0	0	...

Generating the Masking Matrix from the Masks

1	0	0	0	0	0	0	0	...
0	1	0	0	0	0	0	0	...
0	0	1	0	0	0	0	0	...
0	0	0	1	0	0	0	0	...

Generating the Masking Matrix from the Masks

↓ each column represents a pixel

$$H =$$

1	0	0	0	0	0	0	0	...
0	1	0	0	0	0	0	0	...
0	0	1	0	0	0	0	0	...
0	0	0	1	0	0	0	0	...
0	0	0	0	1	0	0	0	...
0	0	0	0	0	1	0	0	...
0	0	0	0	0	0	1	0	...
0	0	0	0	0	0	0	1	...
...								

← each row is a mask

Measuring a Pixel is Matrix-Vector Multiplication

1	0	0	0	0	0	0	0	...
0	1	0	0	0	0	0	0	...
0	0	1	0	0	0	0	0	...
0	0	0	1	0	0	0	0	...
0	0	0	0	1	0	0	0	...
0	0	0	0	0	1	0	0	...
0	0	0	0	0	0	1	0	...
0	0	0	0	0	0	0	1	...
...								

Masking Matrix $\mathbf{H}$

i_1
i_2
i_3
i_n

Unknown,
vectorized
image, $\vec{i}$

=

s_1
s_2
s_3
s_n

Recorded
Sensor
readings, $\vec{s}$

Measuring a Pixel is Matrix-Vector Multiplication

$$\vec{s} = H\vec{l}$$

- We know H and we have the sensor readings, how do we get the image?
- How do we solve this?
- When can we solve this?
 - Conditions on H

Poll Time!

$$\vec{s} = H\vec{i}$$

Select all of the following that must be true for the image vector $\vec{i}$ to be recoverable from the sensor vector $\vec{s}$.

1. H must be invertible
2. H must have linearly independent rows
3. H must be a square matrix
4. H must be the identity matrix

Poll Time!

Select all of the following that must be true for the image vector i to be recoverable from the sensor vector s .

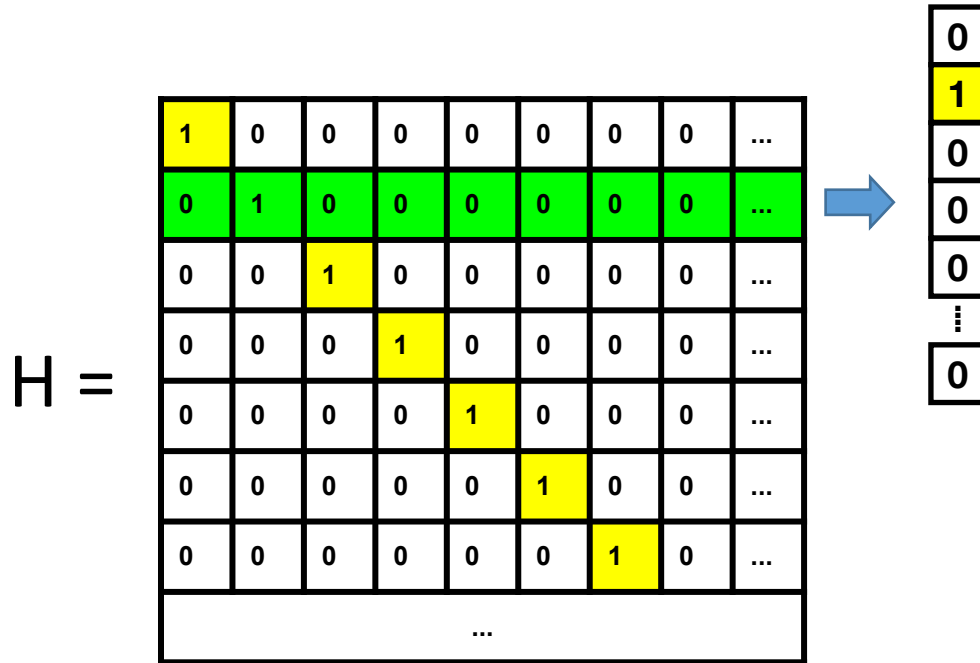
1. H must be invertible
2. H must have linearly independent rows
3. H must be a square matrix
4. H must be the identity matrix

How Scanning Works: iPython

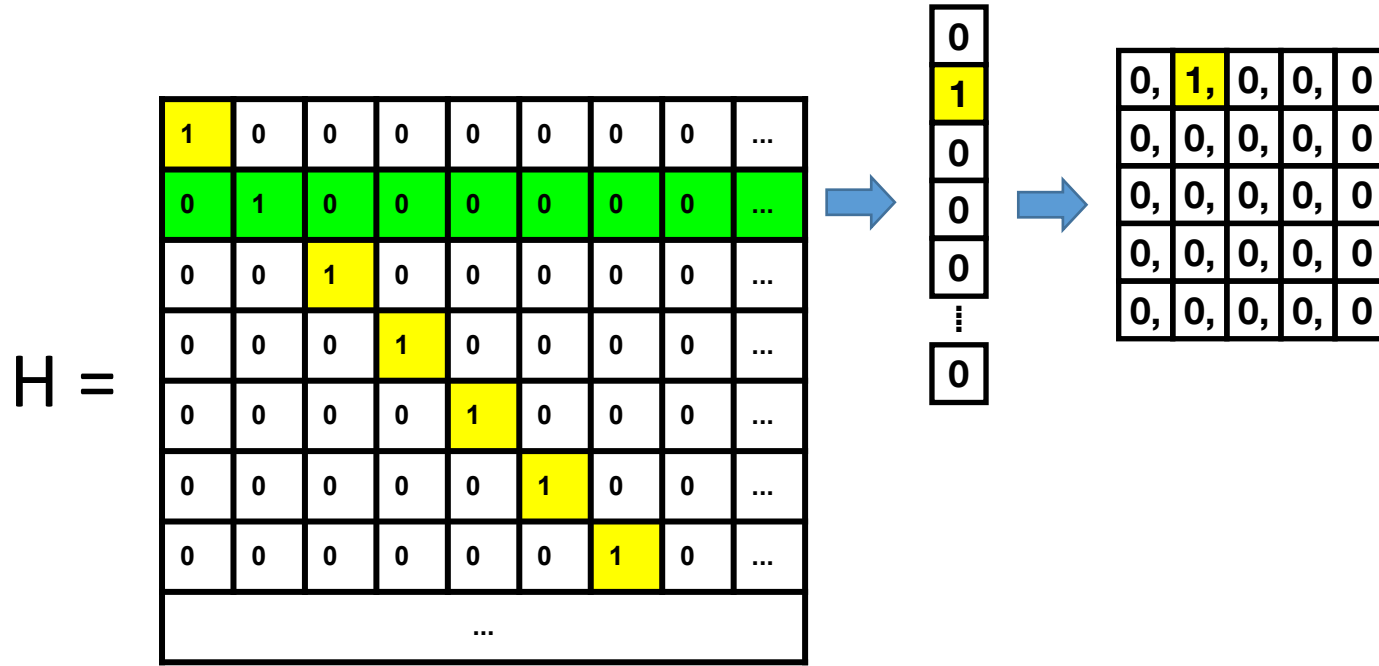
$$H = \begin{matrix} & \begin{matrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots \end{matrix} \\ \begin{matrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{matrix} & \begin{matrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & \dots \end{matrix}$$

...

How Scanning Works: iPython



How Scanning Works: iPython



How Scanning Works in Python

H =

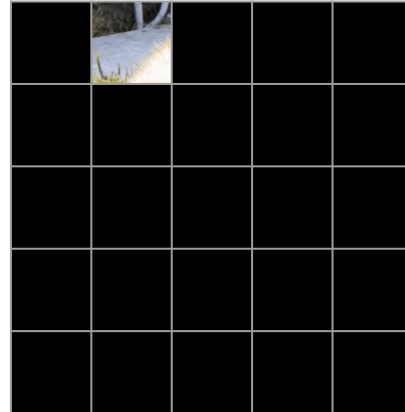
1	0	0	0	0	0	0	0	...
0	1	0	0	0	0	0	0	...
0	0	1	0	0	0	0	0	...
0	0	0	1	0	0	0	0	...
0	0	0	0	1	0	0	0	...
0	0	0	0	0	1	0	0	...
0	0	0	0	0	0	1	0	...
0	0	0	0	0	0	0	1	...
...								



0
1
0
0
0
...
0



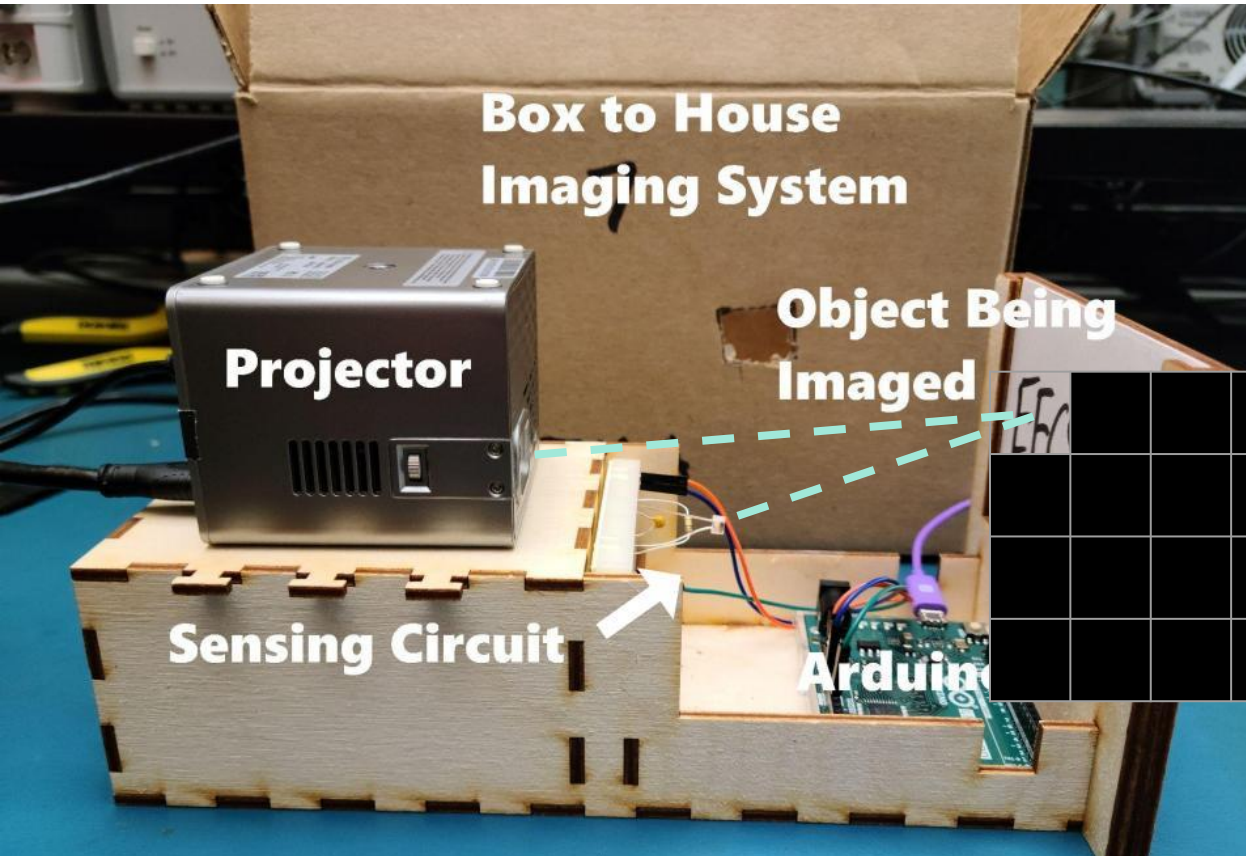
0,	1,	0,	0,	0
0,	0,	0,	0,	0
0,	0,	0,	0,	0
0,	0,	0,	0,	0
0,	0,	0,	0,	0



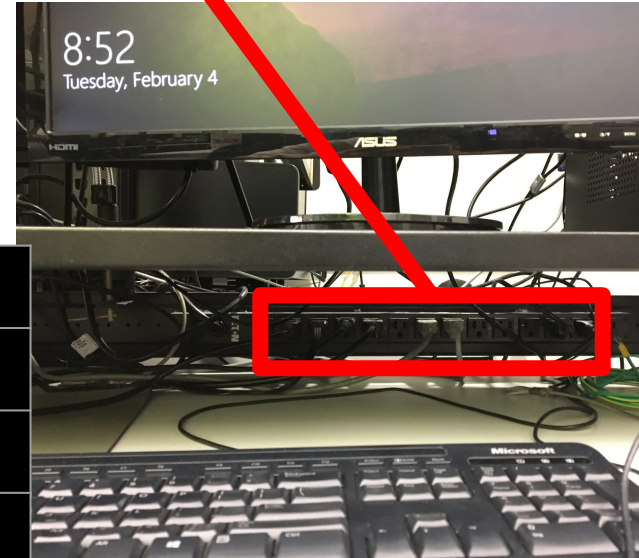
What Makes a Mask Good?

- Linearly independent columns → Invertible
 - Can't get a solution without this
 - There is a unique solution
- What would be a bad mask?
- Food for thought: Are all invertible matrices equally as good?
 - Find out in Imaging 3 next week

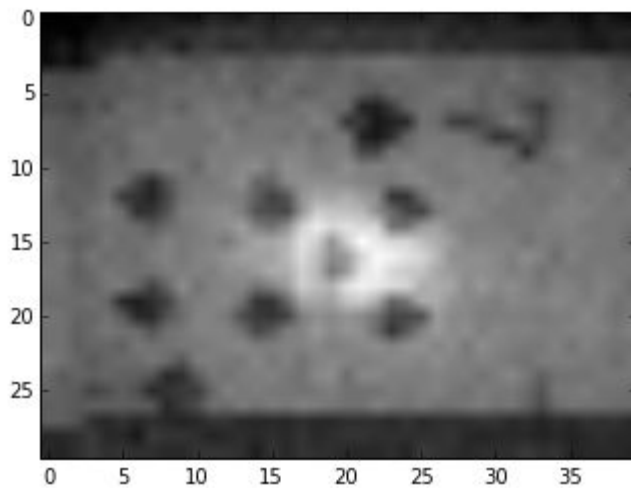
Setup



Power strip to power your projector



Sample Images



Setup

1. Draw a “simple” image
2. Use Python to project masks onto it in a dark environment
3. Measure ambient light sensor reading to get s
4. Multiply by H inverse to find i ($= H^{-1}s$)

Color Imaging!

- The masks we have been using so far have been black and white (1s and 0s). Thus, B&W images
- What if we use color masks instead?
- Make use of RGB (red green blue) channels and reconstruct three different scans
- Same system as before: $\vec{s} = H\vec{i}$
- Only difference is one “system” for each color
- With a bit of math/signal processing, we can get color images!

Tips for a Good Image

- READ CLOSELY. There are many small directions that help you get a good setup
- Focus projector using dial on the side
- Close the box firmly & scan under dark conditions
- Make sure the cables are plugged in, and do not disturb them during the scanning process

Poll Time

Select all of the following that describe the relationship between H (the masking matrix), s (the sensor vector), and i (the image vector)?

1. $Hs = i$
2. $Hi = s$
3. $H^{-1}i = s$
4. $H^{-1}s = i$
5. $i * s = H$

Poll Time

Select all of the following that describe the relationship between H (the masking matrix), s (the sensor vector), and i (the image vector)?

1. $Hs = i$
2. $Hi = s$
3. $H^{-1}i = s$
4. $H^{-1}s = i$
5. $i * s = H$

Debugging



1. Make sure wires/resistors/light sensor are not loose
2. Light sensor orientation: short leg goes into +ve
3. Check COM Port
4. Reupload code to Arduino after making any change in circuit
5. Check Baud Rate in Serial Monitor (115200)
6. Projector might randomly restart in the middle of the lab. Make sure brightness 0 contrast 100.
7. If you see a very bright corner in the scan, move the light sensor away from the projector

Checkoff Form:



Feedback Form:

